A Multiscale Incremental Learning Network for Remote Sensing Scene Classification

Zhen Ye, Yu Zhang, Jinxin Zhang, Wei Li, *Senior Member, IEEE*, and Lin Bai, *Member, IEEE*

***Abstract*— To infer unknown remote sensing scenarios, for remote sensing scene classification (RSSC), most existing deep neural networks (DNNs) are trained on closed datasets. When the acquisition speed and quantity of remote sensing images increase rapidly, these models cannot be used to classify new scenes. Currently, incremental learning as an effective solution for solving the *catastrophic forgetting* issue, but ignoring *the stability-plasticity dilemma*. In this article, we propose a new incremental learning network, named efficient channel attention-based multiscale depthwise network (ECA-MSDWNet), in which efficient channel attention (ECA) improves the model's ability to focus on critical information in complex context, and multiscale depthwise convolution (MSDW Conv) extracts multiscale features in a fine-grained way. In addition, in incremental learning process, we expand new modules based on a dynamic-structure method to fit the residuals between the labels and the outputs of the old model, enhancing the plasticity of the new model for new tasks while maintaining the performance of the old tasks. Finally, we compress the model to reduce redundant parameters and feature dimensions through an effective knowledge distillation strategy. Experiments on four open datasets demonstrate the effectiveness of our method. Our code is available at https://github.com/zhangyu-chd/ECA-MSDWNet.**

***Index Terms*— Catastrophic forgetting, incremental learning, multiscale feature, remote sensing scene classification (RSSC).**

I. Introduction

With the development of deep neural networks (DNNs), remote sensing scene classification (RSSC) has made significant progress in recent years [1], [2], [3]. In the most widely used paradigm, DNN-based models trained on closed datasets are applied to infer unknown remote sensing scenarios. Advances in sensor technology have dramatically increased the speed and volume of remote sensing data acquisition, and there is an urgent need for DNNs that can continually adapt to the addition of new data and learn over time. However, models trained by closed datasets can only be applied to the current task and cannot meet the demand for continuous multitask processing over time. When the model is trained on a new task, its classification performance for the old tasks degrades significantly, which is known as the *catastrophic forgetting* issue [4]. We expect models to be both plastic and stable, with the ability to integrate new knowledge and refine existing knowledge from new data, while eliminating the impact on the performance of the old tasks when learning a new task. The conflicting stability and plasticity constitute *the stability-plasticity dilemma* [5], [6]. The approaches of mixing old and new data to retrain the model, whenever new data are inputted due to reducing the real-time learning capability of the model; at the same time, it requires a large amount of computational resources and storage space, causing a difficulty of deployment on edge computing devices.

Incremental learning is an effective strategy to solve the above problems by continuously training the models with updated data streams and allowing the models to learn new tasks while maintaining the classification of old tasks. Related studies based on incremental learning, focus on the problem of catastrophic forgetting, can be classified into three categories [7], [8]: regularization-based methods, dynamic-structure-based methods, and replay-based methods.

1) Regularization-based methods can be divided into methods based on prior information and methods based on knowledge distillation. Prior information-based methods use estimates of model parameters as a prior when learning a new task and impose a loss function when changing parameters that are important for the old tasks to reduce parameter coverage of the old tasks when learning the new tasks. Representative approaches include the elastic weight consolidation (EWC) [9] and the intelligent synapse (IS) [10]. EWC analyzes the incremental learning framework in probability theory and demonstrates that the second-order derivatives of the parameters can reflect the importance of the model based on the Laplace approximation. IS measures the importance of the parameters by their contribution to the loss decay in an online manner based on EWC. However, the methods based on Fisher matrix are too complex to compute. Pan et al. [11] proposed to use Gaussian process formulation to obtain higher efficiency of the regularization process. Knowledge distillation-based methods extract knowledge from the old network to the new network through the similarity of outputs. Li and Hoiem [12] proposed the learning without forgetting (LwF), which is a representative method based on

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knowledge distillation. LwF adds distillation loss to the loss function, so that the model can retain the features of old tasks when learning new tasks. Wu et al. [13] proposed Bic to mitigate the classification bias caused by imbalance of training data after knowledge distillation by adding a bias correction layer after the fully connected (FC) layer. In summary, the regularization-based approaches protect the learned knowledge by adding additional loss terms to constrain the weights, but this may limit the model's ability to learn new tasks and cannot balance the old and new tasks well. Meanwhile, regularization-based approaches have a large impact on the classification accuracy of the model when the differences between the old and new tasks are large.

2) Dynamic-structure-based approaches generate additional parameters for new tasks in order to reduce mutual interference between parameters of different tasks. One of the most representative approaches is progressive neural networks (PNN) [14], in which each task is associated with a corresponding network and is tuned through lateral links. PNN effectively alleviates the problem of catastrophic forgetting, since it completely avoids the interference of individual subnetworks with task-specific parameters. However, its computational complexity grows exponentially with the number of tasks, making deployment difficult. Xiao et al. [15] proposed a model that grows in a self-organizing manner based on similarity, which allows it to learn faster while inheriting features from the existing models. Yoon et al. [16] proposed dynamically expandable networks (DENs) to expand new neurons from top to down and eliminate useless neurons by group sparsity regularization. Yan et al. [17] proposed dynamically expandable representation (DER) to expand a new backbone network when faced with a new task and to eliminate useless neurons by a larger FC layer to aggregate the functions of the network. Zhou et al. [18] proposed memory-efficient incremental learning algorithm implementation to achieve model expansion with minimal budget cost. The shortcoming of the dynamic-architecture-based approaches is that the number of model parameters increases as the number of tasks.

3) Replay-based approaches argue that various factors limit the access to data of old tasks, resulting in incremental learning that fails to preserve the memory of old tasks. One solution is to generate pseudo-samples, and Shin et al. [19] proposed the deep generative replay (DGR), which uses generative networks to generate pseudo-historical images for the old tasks. Another solution is to store a subset of representative samples from the old tasks and mix them with the new task data for training. Incremental classifier and representation learning (iCaRL) [20] selects the best exemplars for the old tasks to help the model obtain maximum learning efficiency with minimal memory requirements. Liu et al. [21] proposed reinforced memory management (RMM) to save different numbers of exemplars for old tasks according to the task's difficulty. Gradient episodic memory (GEM) [22] uses the stored exemplars to evaluate the update state. Iscen et al. [23] proposed to reduce the storage overhead by keeping the low-dimensional features of the data instead of the old instances.

Recently, incremental learning techniques have received increasing attention and achieved some results in the field of remote sensing image applications. Among them, some results have been achieved in RSSC. Ammour et al. [24] improved the impact on the old land cover classification task by maximizing the distance between tasks. Ye et al. [25] proposed asymmetric collaborative network (SCN) algorithm using asymmetric collaborative networks Pres-Net and Trans-Net for retaining long-term and short-term memory, respectively, and achieving information exchange between the two networks through triple distillation. Lu et al. [26] proposed a lightweight incremental learning network based on a dynamic structure, replacing the role of feature extractor to extract features of a single task instead of sharing features of all tasks to alleviate the backbone, effectively reducing the number of parameters for overcoming catastrophic forgetting. Ammour [27] proposed a memory replay-based approach that uses autoencoder to generate pseudo-images of old tasks to train network with new tasks and analyze the underlying data structure of the tasks.

Although the existing incremental learning methods for RSSC alleviate the catastrophic forgetting to some extent, they still have the following drawbacks. First, some methods based on knowledge distillation may not have enough plasticity to cope with upcoming new tasks due to the single network structure, while the model still suffers from feature degradation of old tasks due to the limited access to old data. Second, dynamic architecture approaches keep expanding new modules with the arrival of new tasks leading to a dramatic increase in the number of parameters, resulting in severe computational and storage overheads that are not suitable for long-term incremental learning tasks. Third, most of these methods do not consider the multiscale features and structural information in remote sensing images (RSIs) [1]. For example, there are many different types of ground objects with significant size variations in RSIs, while many of them are not in the center of the images, which is shown in Fig. 1; it can be seen that the size of ground objects varies greatly in different scenes. These issues have made feature representation learning in RSSC tasks extremely difficult. Therefore, it is important to design multiscale feature extraction networks for the incremental learning task of RSSC.

To solve the above problems, this article proposes a multiscale incremental learning network for RSSC named efficient channel attention-based multiscale depthwise network (ECA-MSDWNet). Among them, the efficient channel attention (ECA) module [28] improves the ability of focusing on key information in complex contexts, and the multiscale depthwise convolution (MSDW Conv) extracts multiscale features by changing the input of convolution. The incremental learning process in this article is based on a dynamic architecture approach that continuously generates new modules to fit the residuals between the labels and the outputs of the model as new tasks come in, maintaining the model's

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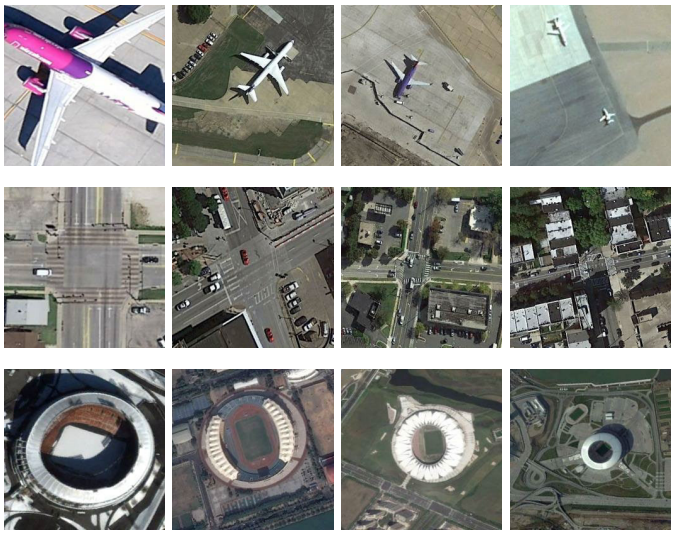


Fig. 1. Multiscale objects of RSI dataset in the following scenes: airplane (the first row), intersection (the second row), and stadium (the third row).

classification performance for old tasks while enhancing the model's plasticity and helping to learn new tasks. At the same time, we lightweight the model by reducing unnecessary parameters and retain key information, avoiding the disadvantage of increasing number of parameters that exists in the dynamic-architecture-based approach.

In summary, the main contributions of our work are as follows.

1) We propose a multiscale incremental learning network for remote sensing scene classification (RSSC). In the proposed network, MSDW conv is designed to extract multiscale features in a fine-grained way, and ECA is employed to improve the model's ability to focus on critical information in complex context.
2) In incremental learning scheme, new modules are added based on dynamic-architecture approaches to fit the residuals between labels and outputs of the old model, allowing the model to maintain the classification for old tasks while enhancing the plasticity for new tasks.
3) We compress the new model to reduce redundant parameters and feature dimensions through an effective knowledge distillation strategy, which avoids the constant growth of the model parameters. Experiments on publicly available datasets demonstrate the effectiveness of our approach, which is promising for RSSC in real-world scenarios.

The remainder of this article is organized as follows. Section II describes in detail the relevant theories of knowledge distillation, dynamic architecture, and replay strategy. Section III describes the proposed ECA-MSDWNet and incremental learning process. Section IV discusses the experimental results. In Section V, we conclude this article.

II. Related Work

In this section, we present incremental learning studies related to the approach of this article: knowledge distillation, dynamic architecture, and replay strategy.

A. Knowledge Distillation

The method based on knowledge distillation aims to encourage the use of teacher model to guide the training of student model, so that the outputs of student model are close to that of teacher model. LwF [12] retains the old model and applies a modified cross-entropy loss function to constrain the model's outputs on the old tasks to retain the ability of old task learning. When learning a new task, the modified cross-entropy loss function is

$$\mathcal{L} = -\lambda_o \sum_{i=0}^{l} y_o'^{(i)} \log \hat{y}_o'^{(i)} - y_n \log \hat{y}_n \tag{1}$$

where l is the number of labels, $y_o'^{(i)}$ and $\hat{y}_o'^{(i)}$ are the modified forms of the old and new model predictions, respectively, and y_n and $\hat{y}_n$ are the true labels and model predictions of the new task, respectively. The former term of the loss function is the distillation loss function, which expects the model's outputs on the old tasks to be similar to the old model, and the latter term is the cross-entropy loss function, which is used to train for new tasks. Knowledge distillation can transfer the knowledge from the more capable teacher network to the less capable student network and enhance the generalization ability of the student network.

B. Dynamic Architecture

Dynamic-architecture-based approaches, represented by PNN [14], generate additional parameters for new tasks. These approaches retain task-specific parameters to avoid knowledge interference between old and new tasks. However, its computational complexity grows exponentially with the number of tasks making practical deployment difficult. Feature boosting and compression for class-incremental learning (FOSTER) [29] analyze the incremental learning from the perspective of gradient boosting, proceduralizing new task learning as a task of feature augmentation. When new data arrive, it freezes the old model F_{t-1} and adds a new module $\mathcal{F}_t$ to fit the residuals between y and output of old model F_{t-1}. Thus, the new model F_t consisting of F_{t-1} and $\mathcal{F}_t$ is able to learn new tasks while maintaining the performance of the old ones. The training process can be expressed as follows:

$$F_t(x) = F_{t-1}(x) + \arg\min_{\mathcal{F}_t} \mathbb{E}_{(x,y)\in\hat{D}_t}\left[\mathcal{L}(y, F_{t-1}(x) + \mathcal{F}_t(x))\right]. \tag{2}$$

Meanwhile, FOSTER argues that not all extended features are necessary for incremental learning, and that the integration of common features to the old and new tasks is required to reduce parameter redundancy. For example, when the old and new tasks contain common features, e.g., basketball and tennis courts have similar features. At this time, forcing the network to extract common features repeatedly is not necessary, so FOSTER transfers the knowledge on the extended network to a single backbone student network through a knowledge distillation strategy, so that the memory does not keep growing with new tasks.

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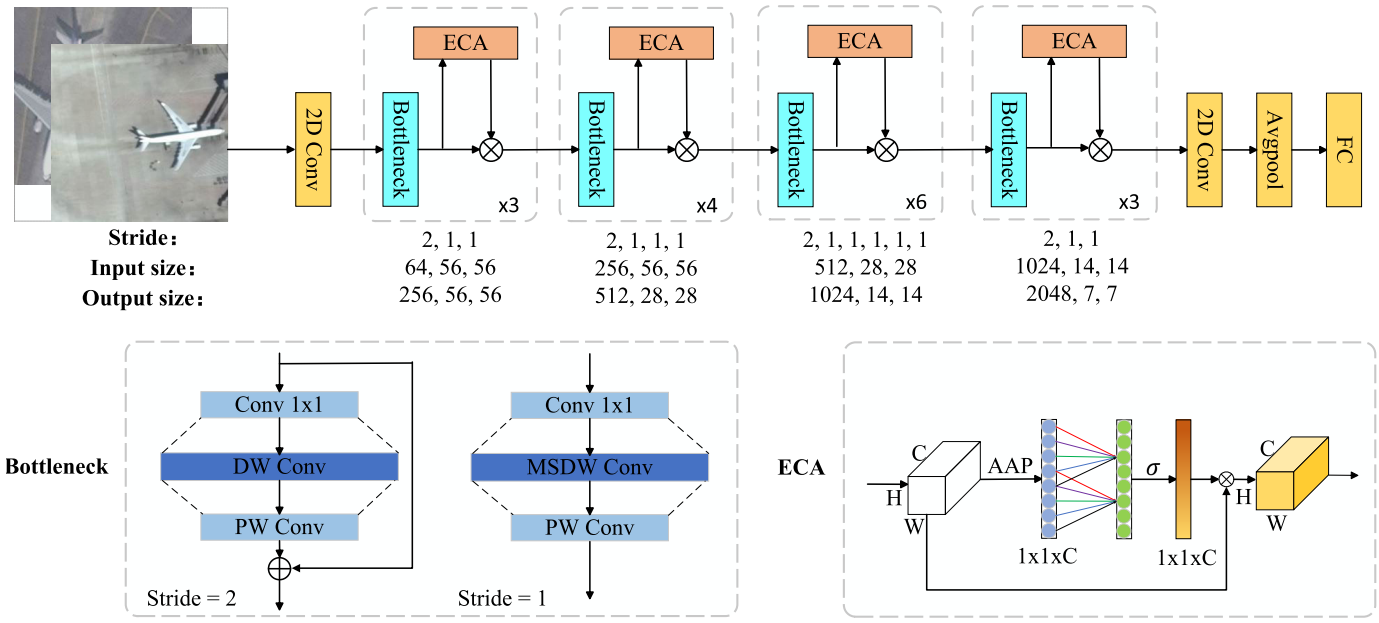


Fig. 2. General structure of ECA-MSDWNet and schematic of bottleneck with attention block. The stride, input, and output size of the bottleneck are listed. When the stride is 1, the bottleneck uses MSDW Conv, and when the stride size is 2, the bottleneck uses DW Conv. AAP: adaptive average pooling.

C. Reply Strategy

Replay strategy enables the model to access some of the data of old tasks. iCaRL [20] allocates a memory to store exemplars from old tasks to be replayed when training a new task. The model keeps a set of exemplars $P \leftarrow (p_1, \ldots, p_m)$ for a certain old class, where m is the number of exemplars. The mean of the features of this exemplar set is closest to the original data and most representative of the class. The selection process is as follows:

$$p_k \leftarrow \underset{x \in X,\ k=1,\ \ldots,m}{\arg\min} \left\| \mu - \frac{1}{k} \left[\varphi(x) + \sum_{j=1}^{k-1} \varphi(p_j) \right] \right\| \tag{3}$$

where x is a sample in a class X and $\mu = (1/|X|)\sum_{x \in X} \varphi(x)$ is the mean of the features in a class.

III. Methodology

A. Incremental Learning Setup

Unlike the traditional case where the model is trained on all classes of available training data. In incremental learning, assume there are N learning phases. In the first phase, we load data $\mathcal{D}_1$ containing training samples of c_1 classes and use $\mathcal{D}_1$ to train the initial classification model (θ_1, w_1), where θ_1 and w_1 denote the parameters of feature extractor and classifier, respectively. When the training in done, we evaluate the model performance on test samples of c_1 classes. Before the second phase, most of the training samples in $\mathcal{D}_1$ are discarded due to the strict memory budget; only a handful of exemplars are preserved in the memory as $\mathcal{V}_1$, which are selected from $\mathcal{D}_1$. In the tth phase ($t \geq 2$), we load all exemplars $\mathcal{V}_{1:t-1} = \mathcal{V}_1 \cup, \ldots, \cup \mathcal{V}_{t-1}$ from the memory and initialize the new model (θ_t, w_t) by the old model (θ_{t-1}, w_{t-1}). We use $\hat{D}_t$ that consists of $\mathcal{V}_{1:t-1}$ and new coming data $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^n$ (containing c_t new classes) to train (θ_t, w_t). Here, n is the number of training samples, $x_i^t \in \mathcal{X}_t$ are the input images, and $y_i^t \in \mathcal{Y}_t$ are the corresponding labels of x_i^t. Then, we evaluate the new model with a test set of all classes $\sum_{j=1}^t c_j$.

B. Overall Architecture

In this article, we propose a multiscale incremental learning network for RSSC. In the network, MSDW Conv is used to extract multiscale features of remote sensing images, ECA achieves local cross-channel interaction and extracts dependencies between channels, and additive model is used for RSIs incremental learning task. In the incremental learning task, differences between old and new tasks can lead to distribution drift. The additive model continuously generates new modules as new data arrives, and the new modules gradually fit the residuals between the labels and the old model's outputs to solve the incremental learning task. However, continuously adding new modules not only leads to an increase in the number of parameters, but also leads to differences between the features of old and new models. For this reason, we compress the boosting model to remove unnecessary parameters and inconsistent features, retaining key information and enhancing the robustness of the model.

The proposed network uses MobileNet V2 as the backbone and replaces the original depthwise convolution (DW Conv) with MSDW Conv, and adds the ECA module to improve the model's performance. The general structures of the network, the bottleneck structure, and the attention module are shown in Fig. 2; the original RSIs are fed into the network as input, and a 2-D convolution is conducted first. Then, bottlenecks and ECA modules are employed for feature extraction. In bottleneck, 1×1 convolution is used to expand the channel dimension of the input, and the MSDW Conv is used to extract multiscale features, and a 1×1 pointwise convolution is utilized to combine the channels to generate new feature map. The ECA achieves local cross-channel interaction and extracts

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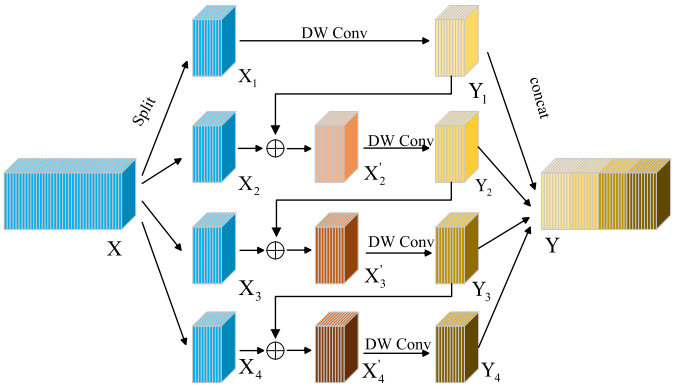


Fig. 3. Structure of MSDW Conv.

interchannel dependencies efficiently by a 1-D convolution on the feature map after adaptive average pooling (AAP). By this local cross-channel interaction strategy without dimensionality reduction, the ECA module improves the network's ability to focus on critical information in complex contexts. Then, the output feature map of ECA is fed into a 2-D convolutional layer, an average pool (Avgpool) layer, and an FC layer to obtain the final classification results.

1) MSDW Conv: Each MSDW Conv splits the input feature map into four partitions along the channel dimension. As an example, Fig. 3 illustrates the operation of an MSDW Conv for an input feature map. That is, in Fig. 3, the input feature map X is partitioned by the split function $\mathcal{S}(\cdot)$; i.e.,

$$X_1, X_2, X_3, X_4 = \mathcal{S}(X). \tag{4}$$

Such that the C channels of X are partitioned into X_1, X_2, X_3, and X_4 with $C/4$ channels each, where C should be divisible by 4. Then, first input partition X_1 is subjected to depthwise convolution to extract features, yielding the output feature map Y_1

$$Y_1 = \mathcal{D}(X_1) \tag{5}$$

where $\mathcal{D}(\cdot)$ denotes the depthwise convolution.

Next, we add Y_1 to the second input partition X_2, and this summation effectively realizes feature reuse. The resulting feature map X_2' is then sent to DW Conv to extract multiscale features. That is,

$$X_i' = Y_{i\text{-}1} + X_i \tag{6}$$

$$Y_i = \mathcal{D}\left(X_i'\right) \tag{7}$$

where $\mathcal{D}(\cdot)$ is depthwise convolution. This process is repeated for $i \in \{2, 3, 4\}$ to process the output feature maps of all partitions. Finally, all the output feature maps are concatenated as follows:

$$Y = \mathcal{C}(Y_1, Y_2, Y_3, Y_4) \tag{8}$$

to yield the final feature map Y, where $\mathcal{C}(\cdot)$ is channelwise concatenation.

In this process, we employ feature reuse to increase the information interaction between different partitions. That is, consider the output Y_1 of the first partition; on the one hand, the output Y_1 of the first partition is added to the input X_2 of the second partition, while, on the other hand, Y_1 is also concatenated to the output of the other partitions to obtain the final output Y; thus, the network uses feature Y_1 multiple times. The MSDW Conv employs similar feature reuse for Y_2 and Y_3 as well.

Furthermore, the MSDW Conv enlarges the receptive field in a multiscale manner. For example, consider the second partition. The output of the first group Y_1 and the input of the second group X_2 are added as the input of the second group X_2'. Because the first partition's output has gone through DW convolution, the receptive field of this part is larger than the input of the second group. The feature maps expand the receptive field in the same way in the third and fourth partition. Thus, MSDW Conv achieves multiscale feature extraction in a fine-grained manner by expanding the receptive field of convolution.

More specifically, traditional multiscale feature extraction methods are often designed as networks of multiple parallel branches, each with a CNN of a fixed kernel size, such that by using different kernel sizes in the different branches, extraction of features with multiple scales is accomplished. In contrast, in the MSDW Conv as proposed here, each partition uses the same CNN kernel size. The multiscale nature of MSDW Conv arises instead from the fact that the output of each CNN is fed into the subsequent partition (i.e., feature reuse), which expands the effective receptive field of the subsequent CNN, thereby providing multiscale feature extraction. Take a cascade of two consecutive 3×3 convolutions as an example: the receptive field of the first convolution is 3×3 (each output value addresses nine values in the input feature map), while the receptive field of the second convolution kernel is effectively 5×5 (each output value addresses 25 values in the original input feature map) when stride $= 1$. The receptive field of the second convolution is larger than that of the preceding convolution even though the kernel itself is the same size; thus, the receptive field is enlarged without the cost of actually using a larger kernel in the CNN. MSDW Conv could, in theory, have as many partitions as channels—this is what makes MSDW Conv much more fine-grained in its multiscale nature.

2) ECA: As a channel attention mechanism, squeeze-and-excitation (SE) network (SENet) [30] pays more channelwise attention to the input feature maps for improving the representative capacity, which is useful for RSSC. The SENet reduces the channel dimensionality of the feature map and captures the relevance of all channels through an FC layer. However, the dimensionality reduction in the SE module has negative effects on the prediction of channel attention, and capturing the relevance of all channels is unnecessary.

Shown as Fig. 2, the ECA module is improved by replacing the FC layer in SE with 1-D convolution after the pooling layer. This strategy helps to preserve more channel information, and local cross-channel information interactions can effectively reduce the number of parameters. The 1-D convolution is implemented as follows:

$$F_{n-1}'^{P} = \mathrm{ECA}(F_{n-1}^{P}) = \mathrm{AAP}(\mathrm{conv1}(F_{n-1}^{P})) \tag{9}$$

where F_{n-1}^{P} denotes the feature map of the input ECA module, conv1 denotes 1×1 convolution, and AAP denotes adaptive average pooling.

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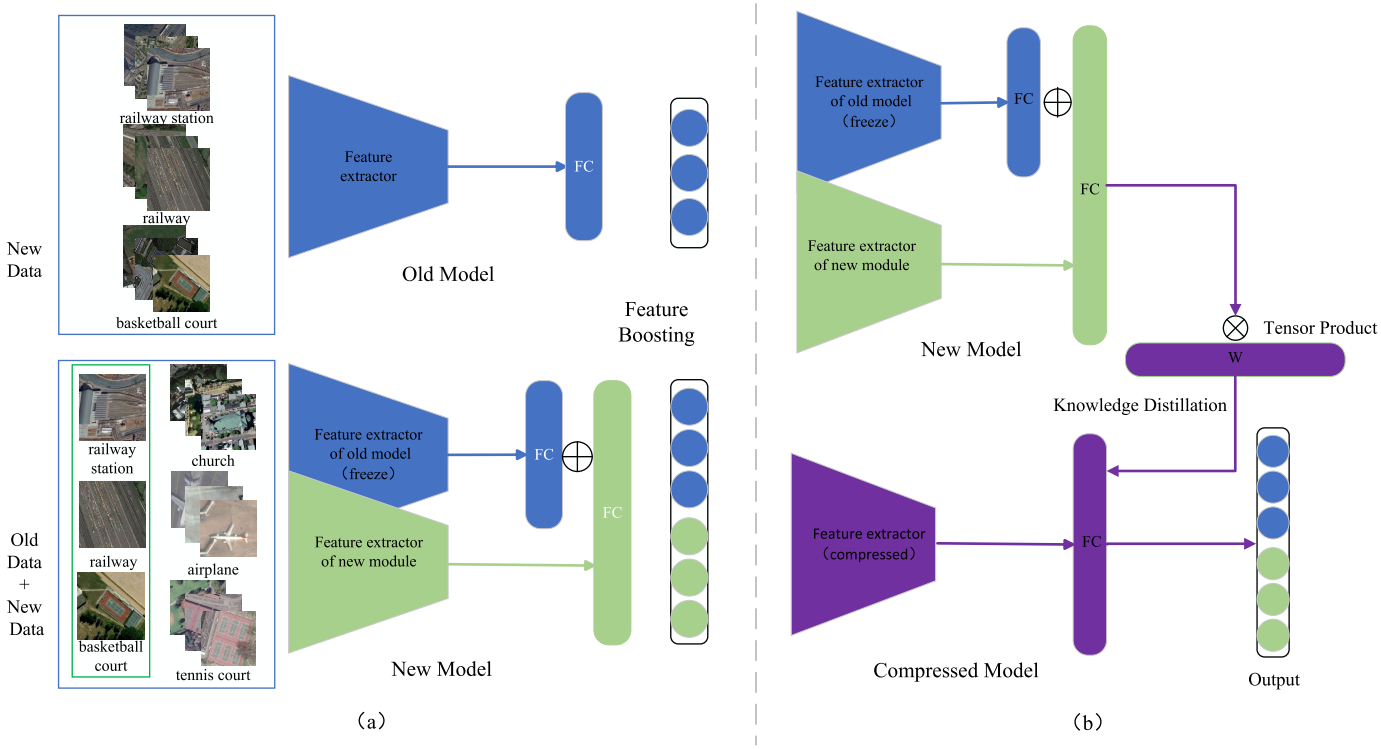


Fig. 4. Incremental learning process: (a) feature boosting and (b) model compression. Feature boosting: when a new task comes, the old model (blue) is frozen. A new module (green) is generated to fit the residuals between the label and the output of old model, which helps the new model learn new classes and old classes better. Compression (b): new model generated in the process of feature boosting is used to instruct the training of the compressed model (purple). Different weights are assigned to the old class and the new class to alleviate the classification bias.

C. Incremental Learning

1) Feature Boosting: Boosting represents a family of machine learning algorithms that convert weak learners to strong ones [31], i.e., combining weak separators $f_i(x)$ to form a strong classifier $F(x)$. We argue incremental learning as a process of feature boosting [29], i.e., incremental learning models continuously learn distinguishing features to classify remote sensing image scenes. The process of feature boosting is shown in Fig. 4(a). When a new task comes, we freeze the old model (blue) and create a new module (green) to fit the residuals between the labels and the outputs of old model. For example, as shown in Fig. 4(a), when the model is trained on a old task consisting of *railway station*, *railway*, and *basketball court*, the **buildings** in the *railway station* are the key feature that distinguishes the *railway station* from the *railway*, and the model may ignore the **lane lines**, which is a common feature to both *railway station* and *railway*. When a new task containing *church*, *tennis court*, and *airplane* arrives, the **lane lines** becomes the key feature that distinguishes the *railway station* from *church*. Therefore, the new model can learn better the features of old and new tasks effectively.

Assume that at the tth incremental learning phase, the old model F_{t-1} of the previous phase is frozen, and F_{t-1} can be further decomposed into feature embedding and linear classifier: $F_{t-1} = (W_{t-1})^\top \Phi_{t-1}(x)$, where $\Phi_{t-1}(\cdot) : \mathbb{R}^D \to \mathbb{R}^d$ and $W_{t-1} \in \mathbb{R}^{d\times|\hat{\mathcal{Y}}_{t-1}|}$, where $|\hat{\mathcal{Y}}_{t-1}|$ is the number of old classes. When a new task arrives, a new module $\mathcal{F}_t$ is trained to fit the residuals of the label y and the output of old model $F_{t-1}(x)$ on $(x, y) \in \hat{D}_t$. The new module $\mathcal{F}_t$ consists of two parts: the feature extractor $\phi_{t-1}(\cdot) : \mathbb{R}^D \to \mathbb{R}^d$ and the linear classifier $\mathcal{W}_t \in \mathbb{R}^{d\times|\hat{\mathcal{Y}}_t|}$, where $|\hat{\mathcal{Y}}_t|$ is the number of all classes. $\mathcal{W}_t$ can be further decomposed into $[\mathcal{W}_t^{(o)}, \mathcal{W}_t^{(n)}]$, where $\mathcal{W}_t^{(o)} \in \mathbb{R}^{d\times|\hat{\mathcal{Y}}_{t-1}|}$ and $\mathcal{W}_t^{(n)} \in \mathbb{R}^{d\times|\hat{\mathcal{Y}}_t|}$. Ideally, the new model $F_t(x)$ satisfies

$$\begin{aligned} y &= F_t(x) = F_{t-1}(x) + \mathcal{F}_t(x) \\ &= \mathcal{S}\left(\begin{bmatrix} W_{t-1}^\top \\ O \end{bmatrix} \Phi_{t-1}(x)\right) + \mathcal{S}\left(\begin{bmatrix} \left(\mathcal{W}_t^{(o)}\right)^\top \\ \left(\mathcal{W}_t^{(n)}\right)^\top \end{bmatrix} \phi_t(x)\right) \end{aligned} \tag{10}$$

where $\mathcal{S}(\cdot)$ is the softmax operation, $O \in \mathbb{R}^{d\times|\mathcal{Y}_t|}$ is the zero matrix, and y is the one-hot vector corresponding to the label y. Denote the parameters of $\mathcal{F}_t$ as θ_t and $\mathrm{Dis}(\cdot,\cdot)$ as the distance metric function, replacing $\mathcal{S}(\cdot) + \mathcal{S}(\cdot)$ with $\mathcal{S}(\cdot + \cdot)$ and replacing $\mathrm{Dis}(\cdot,\cdot)$ with the Kullback–Leibler scatter. This process can be expressed as optimizing θ_t to obtain the optimal solution θ_t^* as follows:

$$\theta_t^* = \arg\min_{\theta_t} \mathrm{KL}\left(y \middle\| \mathcal{S}\left(\begin{bmatrix} W_{t-1}^\top & \left(\mathcal{W}_t^{(o)}\right)^\top \\ O & \left(\mathcal{W}_t^{(n)}\right)^\top \end{bmatrix} \begin{bmatrix} \Phi_{t-1}(x) \\ \phi_t(x) \end{bmatrix}\right)\right). \tag{11}$$

Thus, F_t can be further decomposed into an extended linear classifier W_t and a cascaded feature extractor $\Phi_t(\cdot)$

$$W_t^\top = \begin{bmatrix} W_{t-1}^\top & \left(\mathcal{W}_t^{(o)}\right)^\top \\ O & \left(\mathcal{W}_t^{(n)}\right)^\top \end{bmatrix}, \Phi_t(x) = \begin{bmatrix} \Phi_{t-1}(x) \\ \phi_t(x) \end{bmatrix} \tag{12}$$

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where $W_{t-1}^{\top}$, O, and $\Phi_{t-1}(x)$ are fixed and ϕ_t, $\mathcal{W}_t^{(o)}$, and $\mathcal{W}_t^{(n)}$ are trainable modules. Eventually, the logits of the new model F_t are

$$W_t^{\top}\Phi_t(x) = \begin{bmatrix} W_{t-1}^{\top}\Phi_{t-1}(x) + \left(\mathcal{W}_t^{(o)}\right)^{\top}\phi_t(x) \\ \left(\mathcal{W}_t^{(n)}\right)^{\top}\phi_t(x) \end{bmatrix}. \quad (13)$$

The lower half is the logits of the new classes, and the upper half is the logits of the old classes. The lower half requires the new module $\mathcal{F}_t$ to learn how to correctly classify the new class, thus enhancing the plasticity of the model to compensate for the performance of the new classes. The upper half encourages the new module $\mathcal{F}_t(x)$ to fit y and $F_{t-1}(x)$, thus encouraging $\mathcal{F}_t$ to use more critical features for classification.

When learning a new task, we have an unbalanced training set $\hat{\mathcal{D}}_t = \mathcal{D}_t \cup \mathcal{V}_{1:t-1}$, because only a small number of exemplars were saved from the old tasks, as shown in Fig. 4(a); this imbalance between the number of old and new tasks will result in a classification bias. To alleviate the classification bias, we add a scaling factor γ to the logits of old and new classes, so that the logits during feature boosting are

$$\gamma W_t^{\top}\Phi_t(x) = \begin{bmatrix} \gamma_1\left(W_{t-1}^{\top}\Phi_{t-1}(x) + \left(\mathcal{W}_t^{(o)}\right)^{\top}\phi_t(x)\right) \\ \gamma_2\left(\left(\mathcal{W}_t^{(n)}\right)^{\top}\phi_t(x)\right) \end{bmatrix} \quad (14)$$

where $0 < \gamma_1 < 1$ and $\gamma_2 > 1$, and γ is a diagonal matrix composed of γ_1 and γ_2. By this scaling strategy, the absolute value of logits for old classes is reduced, and the absolute value of logits for new classes is enlarged, thus forcing the model F_t to generate larger logits for old classes and smaller logits for new classes.

The scale factors γ_1 and γ_2 are obtained through the normalized effective number E_n of each class, which can assign larger weights to classes with fewer exemplars. E_n is calculated as follows:

$$E_n = \begin{cases} \dfrac{1-\beta^n}{1-\beta}, & \beta \in [0, 1) \\ n, & \beta = 1 \end{cases} \quad (15)$$

where n is the number of instances and β is an adjustable hyperparameter. Here, we set an adjustable hyperparameter β_1.

Specifically, $(\gamma_1, \gamma_2) = ((E_{n_{\text{old}}}/(E_{n_{\text{old}}} + E_{n_{\text{new}}})), (E_{n_{\text{new}}}/(E_{n_{\text{old}}} + E_{n_{\text{new}}})))$. Therefore, the logits alignment loss $\mathcal{L}_{\text{LA}}$ for balancing the old and new tasks is

$$\mathcal{L}_{\text{LA}} = \text{KL}\left(y \middle\| \mathcal{S}\left(\gamma W_t^{\top}\Phi_t(x)\right)\right). \quad (16)$$

In the process of feature boosting, we added feature enhancement loss $\mathcal{L}_{\text{FE}}$ and knowledge distillation loss $\mathcal{L}_{\text{KD}}$ as supplements to $\mathcal{L}_{\text{LA}}$.

When learning a new task, it is not enough to let the new model $\mathcal{F}_t$ to fit the residuals between the labels y and outputs of $F_{t-1}(x)$. In the extreme case, the residuals between $F_{t-1}(x)$ and y are zero. In that case, the new module $\mathcal{F}_t$ cannot obtain any information about the old classes, and it will impair the performance of the model for the old classes. Therefore, we should prompt the new module $\mathcal{F}_t$ to learn the old classes further. We initialize a new linear classifier $W_t^{(a)} \in \mathbb{R}^{d\times|\hat{\mathcal{Y}}_t|}$, which is used to map the new feature $\phi_t(x)$ to logits of all known classes and require that the new model itself correctly classifies all classes. Thus, even if the residual between $F_{t-1}(x)$ and y is zero, the new feature $\phi_t(x)$ is still able to learn how to classify all the old classes. The feature enhancement loss is

$$\mathcal{L}_{\text{FE}} = \text{KL}\left(y \middle\| \mathcal{S}\left(\left(W_t^{(a)}\right)^{\top}\phi_t(x)\right)\right). \quad (17)$$

Furthermore, it should be noted that simply using a one-hot targets in an unbalanced dataset to train a new feature extractor may result in overfitting to small classes and failing to learn feature representations that generalize well to the old classes. To alleviate this phenomenon and provide more supervision for the old classes, the knowledge distillation is used to encourage $F_t(x)$ to have a similar output distribution to F_{t-1} on the old classes

$$\mathcal{L}_{\text{KD}} = \text{KL}\left(\mathcal{S}(F_{t-1}(x)) \middle\| \mathcal{S}\left(F_{t-1}(x) + \left(W_t^{(o)}\right)^{\top}\phi_t(x)\right)\right). \quad (18)$$

In summary, $\mathcal{L}_{\text{FE}}$ and $\mathcal{L}_{\text{KD}}$ are supplements to $\mathcal{L}_{\text{LA}}$, and the addition of three loss functions jointly acts on the feature boosting process in each incremental learning phase

$$\mathcal{L}_{\text{Boosting}} = \mathcal{L}_{\text{LA}} + \mathcal{L}_{\text{FE}} + \mathcal{L}_{\text{KD}}. \quad (19)$$

2) Model Compression: Incremental learning methods based on dynamic architectures have a disadvantage of adding new modules $\mathcal{F}_t$ to model F_{t-1}, which results in increasing the number of parameters and feature dimensions of model F_t. As shown in Fig. 4(a), when adding a new module $\mathcal{F}_t$ to F_{t-1}, the model F_t becomes a dual-branch structure with an increased number of parameters. At the feature dimension level, *basketball court* in old task and *tennis court* in new task contain common features, such as rectangular shape and lines. In this case, repeated learning of common features is unnecessary. Therefore, we consider compressing the extended feature space of F_t to a smaller space without significantly degrading performance. The process of compression is shown in Fig. 4(b). We remove redundant parameters and reduce dimensions to make the distribution of the same classes more compact. Knowledge distillation is a simple and effective way to achieve this goal. Since our model F_t can handle all visible with excellent performance, it can give a soft target, i.e., the output distribution of all known classes, to any of the inputs in the unbalanced dataset $\hat{\mathcal{D}}_t$. The new model generated in the process of feature boosting in Fig. 4(b) is used to instruct the training of the compressed model.

When training the compressed model $F_t^{(s)}$, in order to reduce the classification bias of the unbalanced training dataset, the class prior should be considered, and the weights of the extracted information of different classes should be adjusted. The balanced knowledge distillation loss function is

$$\mathcal{L}_{\text{BKD}} = \text{KL}\left(w \otimes \mathcal{S}(F_t(x)) \middle\| \mathcal{S}\left(F_t^{(s)}\right)\right) \quad (20)$$

where $\otimes$ denotes the vector product and w is the weighted vector, which is calculated as $(w_{\text{old}}, w_{\text{new}}) = ((E_{n_{\text{old}}}/(E_{n_{\text{old}}} + E_{n_{\text{new}}})), (E_{n_{\text{new}}}/(E_{n_{\text{old}}} + E_{n_{\text{new}}})))$. E_n is shown

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in (15), and here, we set an adjustable hyperparameter β_2 to control E_n.

Algorithm 1 Pseudocode for Our Method Training (tth Phase, $t \geq 2$)

Input: New data $\mathcal{D}_t$; exemplars of old data $\mathcal{V}_{0:t-1}$; last-phase old model F_{t-1} with (θ_{t-1}, w_{t-1}); hyperparameters β_1 and β_2.

Output: Compressed model perform well on all known classes.

Feature Boosting
Initialize new module $\mathcal{F}_t$ with (θ_{t-1}, w_{t-1});
for *epochs* **do**:
    Train $\mathcal{F}_t$ using $\hat{D}_t = \mathcal{D}_t \cup \mathcal{V}_{0:t-1}$;
    Calculate $\mathcal{L}_{LA}$, $\mathcal{L}_{FE}$, $\mathcal{L}_{KD}$;
    $\mathcal{L}_{Boosting} = \mathcal{L}_{LA} + \mathcal{L}_{FE} + \mathcal{L}_{KD}$;
    Update $\mathcal{F}_t$
end for
Old model F_{t-1} and new module $\mathcal{F}_t$ form new model F_t
Model Compress
Initialize compressed model (θ_t, w_t) with (θ_{t-1}, w_{t-1})
for *epochs* **do**:
    Train (θ_t, w_t) using $\mathcal{D}_t \cup \mathcal{V}_{0:t-1}$;
    Calculate $\mathcal{L}_{BKD}$,
    Update (θ_t, w_t) of compressed model;
end for
Retain the compressed model as F_t with (θ_t, w_t);
Select exemplars $\mathcal{V}_t$ from $\mathcal{D}_t$.

3) Algorithm Scheme: When learning a new task, the pseudocode at the tth phase ($t \geq 2$) is shown in Algorithm 1. When a new task $\mathcal{D}_t$ arrives, last-phase model F_{t-1} is frozen, and new module $\mathcal{F}_t$ is added. $\mathcal{F}_t$ is trained on $\hat{D}_t = \mathcal{D}_t \cup \mathcal{V}_{0:t-1}$, and the loss function of training $\mathcal{F}_t$ is $\mathcal{L}_{\text{Boosting}} = \mathcal{L}_{\text{LA}} + \mathcal{L}_{\text{FE}} + \mathcal{L}_{\text{KD}}$. F_{t-1} and $\mathcal{F}_t$ form a new model F_t. To remove redundant parameters and reduce feature dimensions, we compress the new model. The parameters of the compressed model (θ_t, w_t) are initialized with (θ_{t-1}, w_{t-1}). The loss function of model compression is $\mathcal{L}_{\text{BKD}}$. The compressed model is used as the final model F_t to classify the test data. Finally, we select exemplars $\mathcal{V}_t$ from $\mathcal{D}_t$ by herding [20].

IV. Experiments and Results

A. Experimental Setting

We first compare the classification performance of our method with that of some classical methods in incremental scenarios using three benchmark datasets, such as the UC-Merced land-use dataset (UC-Merced) [32], the Aerial Image dataset (AID) [33], and the Remote Sensing Image Classification Benchmark (RSICB-256) [34]. Then, we analyze the changes in model performance as the number of tasks growing, the role of each module in the method using the Northwestern Polytechnical University (NWPU) Remote Sensing Image Scene Classification (NWPU-RESISC45) dataset [35] and the robustness of our method to hyperparameters in ablation experiments. Specifically, we set up four different incremental learning scenarios, three tasks with seven classes for UC-Merced, six tasks with five classes for AID, nine tasks with four classes for RSICB-256, and five tasks with nine classes for NWPU.

1) Datasets: We selected four remote sensing datasets for the experiments. The UC-Merced, AID, and RSICB-256 were used for the comparison of the algorithms. In addition, the NWPU-RESISC45 dataset [35] was used for ablation experiments.

1) The UC-Merced land-use dataset consists of 21 types of scene classes, and it is abbreviated as UC-Merced in this article. Each scene class consists of 100 images with 256 × 256 pixels. The spatial resolution of each image is 0.3 m. The UC-Merced is a representative and small dataset.
2) The AID consists of 30 types of scene classes. Each scene class consists of 220–420 images with 600 × 600 pixels. The spatial resolution of each image is between 0.3 and 8 m. The AID is a representative, inhomogeneous, and difficult dataset.
3) The RSICB-256 consists of 35 types of scene classes. Each scene class consists of 800 images with 256 × 256 pixels. The spatial resolution of each image is between 0.22 and 3 m. The RSICB-256 is a complex dataset containing mislabeled noise.
4) The NWPU-RESISC45 dataset consists of 45 types of scene classes. Each scene class consists of 700 images with 256 × 256 pixels. The spatial resolution of each image is between 0.2 and 30 m. The NWPU is a multiscale dataset. Furthermore, the NWPU dataset was utilized for analyzing the effect of the MSDW Conv.

2) Baselines: Seven classical incremental learning methods were employed to compare with our approach: **Joint**, **Finetune**, **EWC** [9], **LwF** [12], **iCaRL** [20], **RMM** [21], and **SCN** [25]. **Joint** is the ideal state method that trains all data together and avoids catastrophic forgetting. **Finetune** trains the new tasks directly on the old model, an approach that does not have any countermeasures against catastrophic forgetting and ignores the performance of the old tasks. **EWC** is a method based on parameter regularization, which calculates the importance of parameters by Fisher matrix. **LwF** is the most typical method based on knowledge distillation, which adds the distillation loss to the loss function, so that the model can retain the features of old tasks. **iCaRL** calculates the sample centers for each class, classifies the test images according to the sample centers, and saves a certain number of samples for old classes to be used for training when a new task comes. **RMM** is a approach based on data replay, where the number of samples saved for each task is selected based on the difficulty of the task. **SCN** proposes an asymmetric collaborative network, and Pres-Net and Trans-Net are used for preserving long-term and short-term memory, respectively, and enable information exchange between the two networks through triple distillation.

3) Evaluation Metrics: Similar to [36], the model is evaluated on all previously learned tasks after training each new task. We use three metrics to evaluate the performance of the

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TABLE I
MEAN ACCURACY FOR DIFFERENT METHODS

Number of sub-tasks	mACC(%) Joint	Finetune	EWC	LwF	iCaRL	RMM	SCN	Ours
3(UC Merced)	**93.33**[1]	41.19	37.38	53.57	84.76	70.95	**90.47**[3]	**94.29**[2]
6(AID)	**99.37**[1]	21.50	27.40	42.95	61.70	61.60	**85.98**[3]	**86.75**[2]
9(RSICB-256)	**99.37**[1]	11.97	14.62	19.67	85.78	**88.42**[3]	87.33	**91.10**[2]

TABLE II
BWT FOR DIFFERENT METHODS

Number of sub-tasks	BWT(%) Joint	Finetune	EWC	LwF	iCaRL	RMM	SCN	Ours
3(UC Merced)	-	75.71	74.28	55.00	**8.92**[2]	15.71	**14.29**[3]	**3.93**[1]
6(AID)	-	88.42	83.62	55.27	41.72	**29.11**[3]	**9.36**[1]	**11.94**[2]
9(RSICB-256)	-	96.02	83.34	64.78	14.92	**11.67**[2]	**13.85**[3]	**5.91**[1]

algorithm, including: accuracy, mean accuracy (mACC) [22], and backward transfer (BWT) [36].

a) Accuracy: Accuracy is the ratio of the number of correct predictions to the total number of predictions and is used to evaluate the performance of the algorithm on a single task. It can be calculated by

$$\text{accuracy}(T) = \frac{\text{Num}(\hat{y} = y)}{\text{Num}(y)} \tag{21}$$

where T denotes the task, $\hat{y}$ denotes the output of the model on the Tth task, and y denotes the true label.

b) mACC: mACC is the mean accuracy of the model on all tasks after learning the last task, which reflects the model's ability to balance learning new tasks and retaining knowledge of old tasks. The larger the mACC, the more effective the incremental learning model is. It can be calculated as follows:

$$\text{mACC}(i) = \frac{1}{T}\sum_{i=1}^{T} P_{T,i} \tag{22}$$

where T denotes the total number of tasks learned by the model and $P_{T,i}$ denotes the accuracy of the model on the ith task after learning the Tth task.

c) BWT: The BWT is used to measure the forgetting of the old task by the model after learning the new task. It can be calculated by

$$\text{BWT}(i) = \frac{1}{T-1}\sum_{i=1}^{T-1}\left(P_{i,i} - P_{T,i}\right). \tag{23}$$

When BWT > 0, it means that the model learns the new task when the performance of the old tasks is affected, i.e., the model faces forgetting. On the contrary, when BWT < 0, it indicates that the model's performance on the old task is improved when it is trained on the new task. Therefore, the smaller the BWT, the less the model forgets the old tasks.

4) Training Details: In our experiments, ResNet-18 was used as the backbone network for all comparison experiments. The training and test sets were divided at a ratio of 4 : 1. The hyperparameters of the comparison methods are set as follows, $T = 2$ in LwF, $\lambda = 15$ in EWC, and the size of exemplar set is 200 in iCaRL and RMM. In our method, $\beta_1 = 0.88$, $\beta_2 = 0.89$, $T = 2$, and the size of exemplar set is 200. All experiments were performed using stochastic gradient descent (SGD) strategy with an initial learning rate of 0.1, learning rate variation using cosine annealing strategy with momentum 0.9, and the weight recession is 1×10^{-5}. The number of initial training epochs for our method is 200, the number of feature boosting epochs is 170, and the number of compression epochs is 130. The number of epochs in each phase for SCN method is 200. For other comparison methods, the number of epochs in initial phase is 200, and the number epochs in remaining phases is 170. All methods in the experiments are initialized with kaiming_uniform. All experiments were performed on an RTX3090Ti GPU operating in Python 3.8 and PyTorch 1.9.1.

B. Results and Analysis

We compare our approach with several state-of-the-art methods, and Tables I and II show the results of mACC and BWT metrics, respectively. For evaluating the capability of the mentioned incremental learning algorithms, we conducted experiments using three datasets. The small dataset UC-Merced was divided into three tasks, the AID was divided into six tasks, and the RSICB-256 was divided into nine tasks.

According to Table I, the joint method has the highest mACC values on the three datasets, because joint learns all the data simultaneously, thus avoiding catastrophic forgetting. Finetune directly trains the model on the new tasks, ignoring the model's performance on the old tasks, which leads to catastrophic forgetting. EWC limits the update of the model's parameters during learning a new task based on parameter regularization, thus effectively preserving the model's performance on the old task, but its mACC gains only a small improvement compared to finetune. LwF and iCaRL use the strategy of knowledge distillation and also obtain higher mACC values, indicating that the knowledge distillation strategy has good properties in incremental learning. ICaRL saves a set of exemplars for each old class and obtained a higher mACC value than LwF, which also indicates the effectiveness of replaying historical data. SCN achieves incremental learning by saving long-term and short-term memories,

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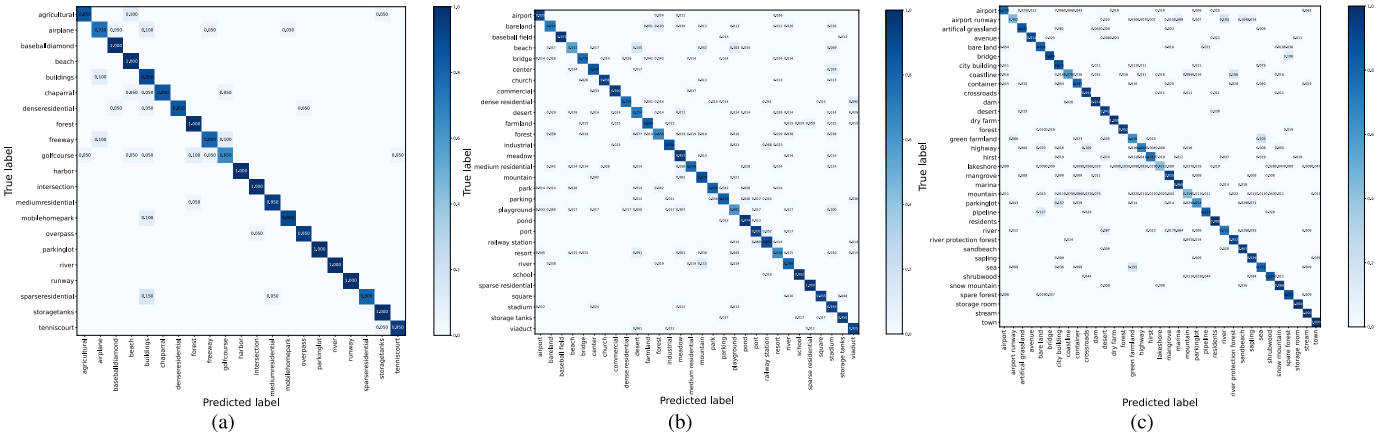


Fig. 5. Confusion matrixes on our methods. (a) UC-Merced. (b) AID. (C) RSICB-256.

separately, and yields high mACC values in all three datasets. Our method continuously generates new modules to fit the residuals between the labels and the old model outputs as new tasks come, maintaining the model performance on the old tasks while making the model plastic enough to learn new data. Our method got the first place in mACC among all incremental learning methods (except Joint), indicating the effectiveness of our method on incremental learning tasks. Fig. 5 shows the confusion matrices of the classification results of the proposed method on the three datasets UC-Merced, AID, and RSICB-256.

The BWT reflects the forgetting of the old tasks when the model learns new tasks. According to the results shown in Table II, our method is minimal on the UC-Merced and RSICB-256 dataset, illustrating the strong ability of our method in combating catastrophic forgetting. For the AID dataset, the BWT value of our method is 2.58% higher than that of SCN, which may be caused by the uneven number of individual scene classes in the AID dataset. This inhomogeneity causes the model parameters to favor the class with a higher number of images, causing the model to produce classification bias in incremental learning. In addition to, there is no BWT value, because joint trains all available data simultaneously. Finetune has no countermeasures against catastrophic forgetting, so it has the largest BWT value, indicating that it has the most severe forgetting of old tasks. Compared to Finetune, EWC and LwF can play a role in limiting the variation of model parameters and preserving old knowledge, but forgetting is still severe. iCaRL and RMM have better performance in terms of BWT values, indicating that the strategy of preserving exemplars for old tasks is effective in incremental learning. SCN also achieves good performance by preserving long-term and short-term memory, respectively.

Table III shows the accuracy of SCN and ours method on each task after learning all tasks sequentially on there datasets. On UC-Merced, AID, and RSICB-256, the accuracy of our method is higher than SCN on two, three, and six tasks, respectively, which indicates the superiority of our method compared to SCN on incremental learning tasks. Also, the accuracy of our method exceeds 80% on all tasks, which indicates that our method can learn new tasks while maintaining the performance on old tasks well and can alleviate the stability-plasticity dilemma better.

TABLE III
ACCURACY (%) OF EACH TASK

		1	2	3	4	5	6	7	8	9
UC Merced	SCN	85.00	86.42	100						
	Ours	**87.86**	**97.14**	97.86						
AID	SCN	71.81	84.61	92.72	88.23	82.92	95.58			
	Ours	**81.72**	81.18	85.62	85.88	**89.22**	**97.62**			
RSICB-256	SCN	79.17	90.96	80.56	91.89	83.38	99.61	82.55	78.71	99.06
	Ours	**87.75**	85.78	**95.25**	**93.98**	**90.11**	83.71	**93.81**	**93.09**	97.45

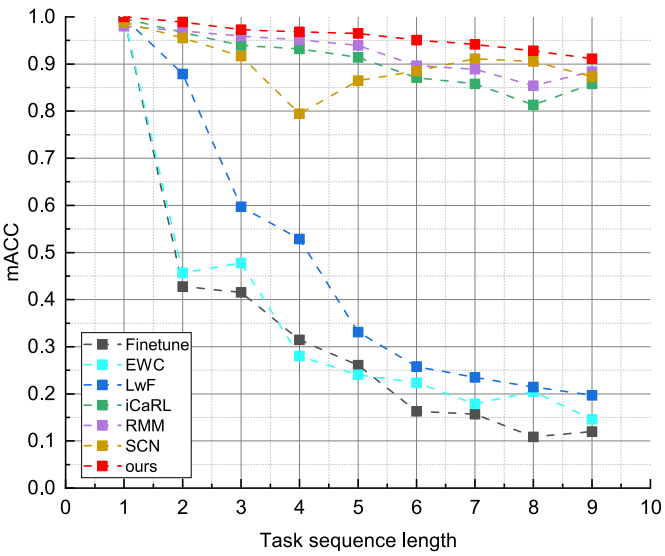


Fig. 6. Performance with sequence length. mACC versus the task sequence length.

C. Sequential Learning Analysis

Generally, the classification performance of each method decreases, as the number of tasks increases. As an example, the change in mACC for each method as the number of tasks increases on the nine tasks of the RSICB-256 dataset is shown in Fig. 6. Our method maintains the mACC metric at a stable and high level, although the number of tasks is high.

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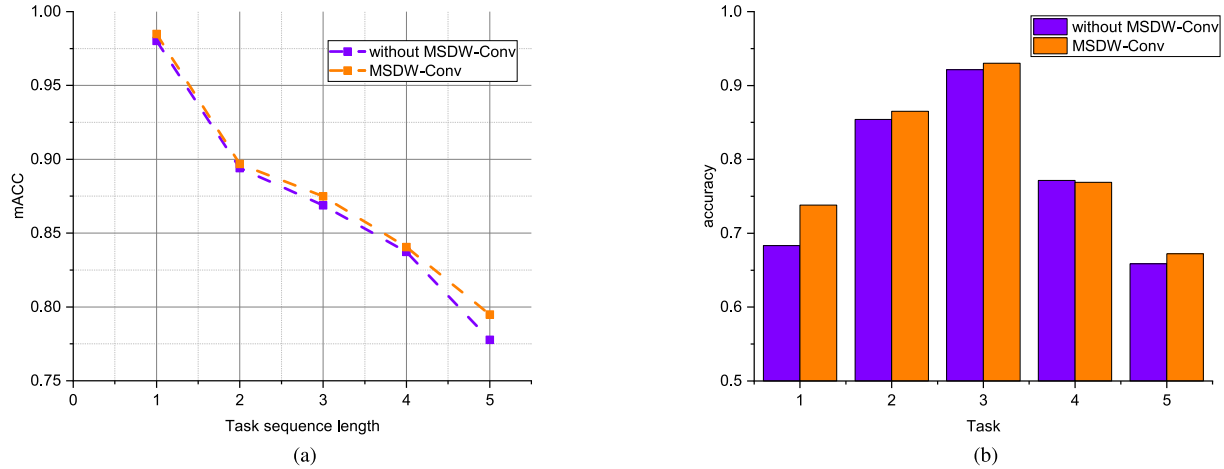


Fig. 7. Feature extraction ablation experiments. The MSDW Conv ablation experiments were performed under a five-tasks NWPU experimental setup. (a) and (b) Comparison of mACC and accuracy without and with MSDW Conv, respectively.

TABLE IV
ABLATION EXPERIMENTS FOR MSDW CONV

	UC Merced		AID		RSICB-256	
	mACC(%)	BWT(%)	mACC(%)	BWT(%)	mACC(%)	BWT(%)
Without MSDW Conv	92.38	4.29	85.55	13.40	90.45	9.18
MSDW Conv	**94.29**	**3.93**	**86.75**	**11.94**	**91.1**	**5.91**

TABLE V
ABLATION EXPERIMENTS FOR ECA

	UC Merced		AID		RSICB-256	
	mACC(%)	BWT(%)	mACC(%)	BWT(%)	mACC(%)	BWT(%)
Without ECA	91.43	5.36	84.90	14.56	89.09	6.55
ECA	**94.29**	**3.93**	**86.75**	**11.94**	**91.10**	**5.91**

For comparable methods, the mACC of finetune decreases sharply with the number of tasks, which indicates that DNNs have a catastrophic forgetting problem and that the model cannot scale to new tasks if nothing is done. As for the EWC and LwF, their mACC drop is not as sharp as finetune's, but the performance illustrates that simple models have limitations in overcoming catastrophic forgetting in incremental learning tasks in the face of increasing number of tasks. The mACC values of iCaRL and RMM decrease slowly with the increase of the number of tasks, indicating the effectiveness of the replay strategy in incremental learning tasks. Meanwhile, RMM saves different number of examplars according to the difficulty of the task and obtains better results than iCaRL by this strategy.

D. Ablation Experiments

In this article, we argue that our method can effectively extract RSIs multiscale features and alleviate catastrophic forgetting because of the following: 1) the MSDW Conv; 2) the ECA module; 3) the combined loss function; and 4) the model compression.

1) MSDW Conv: We conducted ablation experiments of MSDW Conv for three datasets, and the results are shown in Table IV. According to the results, the proposed network achieves higher mACC value and the lower BWT value compared with the network without MSDW Conv, indicating that MSDW Conv is effective for incremental learning. To further demonstrate the role of MSDW Conv, we also conducted ablation experiments on the NWPU dataset. The NWPU consists of scene images with various sizes of ground objects. Fig. 7(a) and (b) shows the comparison of mACC and accuracy of the models without and with the MSDW Conv on the five-tasks NWPU dataset, respectively. Fig. 7(a) shows that the mACC values of the model with MSDW Conv are higher than those without MSDW Conv at each phase of incremental learning. Fig. 7(b) shows the accuracy on each task after learning all tasks. The accuracy that with MSDW Conv has only a slight decrease in on the fourth task. The accuracies on the remaining four tasks were higher than those of the model without MSDW Conv, by 5.48%, 1.11%, 0.88%, and 1.35%, respectively. The above experimental results indicate that the multiscale feature extraction module MSDW Conv can effectively improve the classification performance of incremental learning on remote sensing scenes with large changes in scene scale. The comparison of the confusion matrices is shown in Fig. 8, where the method with MSDW Conv improves the accuracy of classification in most scene classes compared to the one without MSDW Conv.

2) ECA Module: We conducted ablation experiments for ECA modules for three datasets, and the results are shown in Table V. The proposed network with the ECA modules achieves higher mACC values and lower BWT values. To better understand the role of the ECA modules, Fig. 9(a) and (b) show the comparison of mACC and accuracy of methods without and with the ECA modules on the three-tasks UC-Merced dataset, respectively. The results in Fig. 9(a) show that the ECA module improves the mACC by 2.14%, 3.5%, and 2.86% at the three phases, respectively. Fig. 9(b) shows the accuracy on each task after learning all the tasks, and the model with the ECA module improves accuracy by 0.72%, 5.71%, and 2.15% on the three tasks, respectively, compared to the model without ECA. The above experimental results indicate that

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TABLE VI
ABLATION EXPERIMENTS FOR LOSSES OF FEATURE BOOSTING

Losses			UC Merced		AID		RSICB-256	
$\mathcal{L}_{LA}$	$\mathcal{L}_{FE}$	$\mathcal{L}_{KD}$	mACC(%)	BWT(%)	mACC(%)	BWT(%)	mACC(%)	BWT(%)
✓			90.71	6.43	81.65	15.70	89.62	10.98
✓	✓		93.10	6.07	84.90	14.29	91.02	10.19
✓		✓	91.19	5.36	84.45	12.67	90.67	8.03
✓	✓	✓	**94.29**	**3.93**	**86.75**	**11.94**	**91.10**	**5.91**

TABLE VII
EFFECT OF MODEL COMPRESSION

	UC Merced (3 tasks)			AID (6 tasks)			RSICB-256 (9 tasks)		
	Parameters	mACC(%)	BWT(%)	Parameters	mACC(%)	BWT(%)	Parameters	mACC(%)	BWT(%)
With compression	**11.19M**	**94.29**	**3.93**	**11.22M**	**86.75**	**11.94**	**11.23M**	**91.10**	**5.91**
Without compression	22.39M	80.71	21.07	55.99M	78.05	22.64	89.59M	71.69	29.31

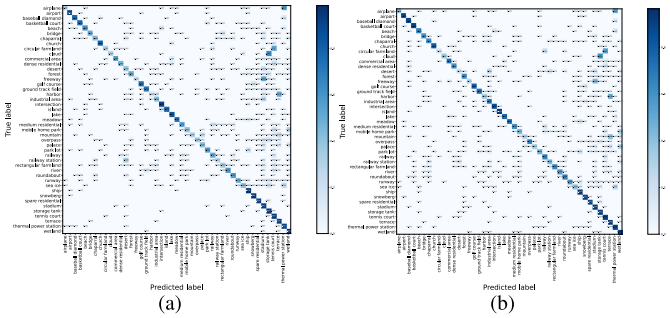


Fig. 8. Comparison of CMs on the NWPU dataset. (a) Without MSDW Conv. (b) MSDW Conv.

the ECA module can enhance the model's ability to focus on critical information. The comparison of the confusion matrices is shown in Fig. 10, and the models with ECA outperformed those without ECA in most classes.

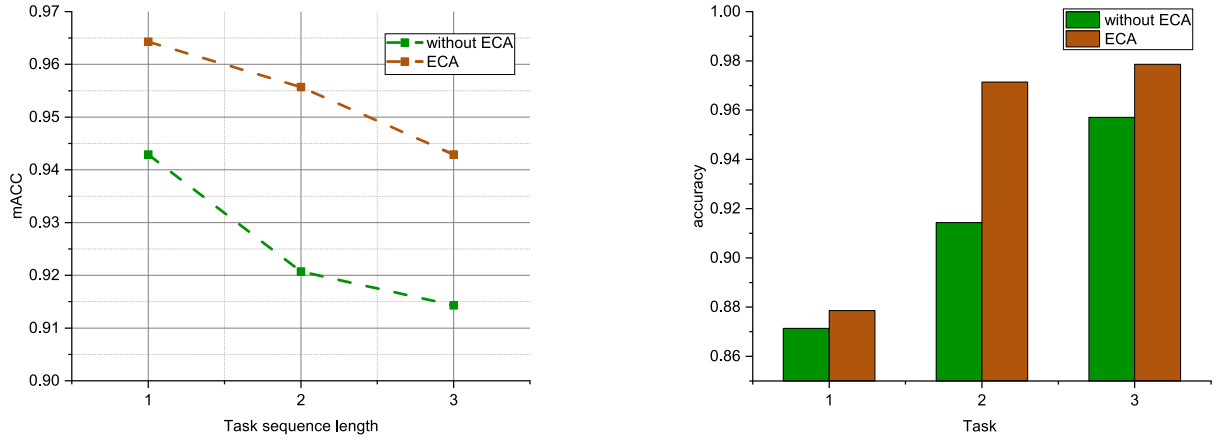
3) Combined Distillation Loss Functions: Our method combines three loss functions $\mathcal{L}_{\text{LA}}$, $\mathcal{L}_{\text{FE}}$, and $\mathcal{L}_{\text{KD}}$, in the feature boosting process. $\mathcal{L}_{\text{LA}}$ enables the model to learn new and old classes by fitting the residuals between the labels and the outputs of the old model, which can also alleviate the classification bias caused by the imbalance of the old and new classes. $\mathcal{L}_{\text{FE}}$ and $\mathcal{L}_{\text{KD}}$ are supplements to $\mathcal{L}_{\text{LA}}$, where $\mathcal{L}_{\text{FE}}$ enables the new model to learn all known classes when the residual is small, and $\mathcal{L}_{\text{KD}}$ transfers the knowledge of the old model to the new model through knowledge distillation. The addition of the three loss functions jointly acts on the feature boosting process of each incremental learning phase. We conducted ablation experiments of the combined loss on three datasets, and the results were shown in Table VI. The experimental results show that the highest mACC value and the lowest BWT value can be obtained by using the three loss functions simultaneously in our method. Fig. 11 shows the results on the three tasks of the UC-Merced dataset. According to results, the mACC is lowest when using $\mathcal{L}_{\text{LA}}$ alone, indicating that relying on the new module alone to fit the residuals between the old model outputs and the labels is insufficient. After adding $\mathcal{L}_{\text{FE}}$ and $\mathcal{L}_{\text{KD}}$, respectively, the mACC increases by 2.39% and 1.19% compared to $\mathcal{L}_{\text{LA}}$ alone, indicating that the strategy of conbined loss is effective. Comparably, our method achieves the highest mACC value using the combined loss functions, improving by 3.58%, 1.19%, and 2.39%, respectively.

4) Model Compression: For our method, not all extended features are necessary, and redundant parameters may make the model complex, so we compress the model by knowledge distillation. To demonstrate the usefulness of compression, we conducted experiments for the proposed method with or without the model compression for three datasets. According to the results in Table VII, the number of parameters will increase proportionally and dramatically without model compression due to incremental learning. Meanwhile, compressed model achieves higher mACC value and lower BWT value for three datasets, which indicates that the compressed model is beneficial to RSSC tasks in incremental learning. As for the reason, we argue that when the training epochs and the amount of training data are stable, too many parameters may make the problem of underfitting. Thus, the ablation experiment proves that it is necessary to perform model compression for our method.

E. Effect of Hyperparameters

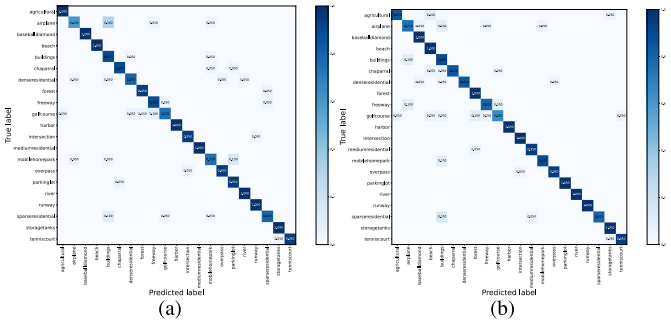
Our method involves two hyperparameters: β_1 and β_2. β_1 alleviates the classification bias caused by the imbalance between old and new classes in the feature boosting process. β_2 adjusts the weights of features extracted from different

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(a) (b)

Fig. 9. Feature extraction ablation experiments. The ECA ablation experiments were performed under a three-tasks UC-Merced experimental setup (a) and (b) comparison of mACC and accuracy without and with the ECA module, respectively.



(a) (b)

Fig. 10. Comparison of CMs on the UC-Merced dataset. (a) Without ECA. (b) ECA.

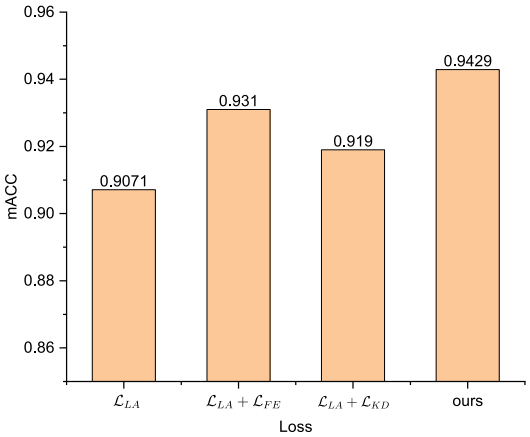


Fig. 11. Loss function of feature boosting.

classes in the model compression process. In order to analyze the effects of β_1 and β_2 on the classification performance of the proposed model, we conducted the following experiments. First, we found that the classification accuracies is satisfied when β_1 and β_2 vary from 0.84 to 0.91. Therefore, we conducted detailed experiments in this range, where the steps of β_1 and β_2 are both 0.01. As shown in Fig. 12, the mACC value is optimal when β_1 is 0.88 and β_2 is 0.89 for UC-Merced. For AID and RSICB-256, we use the same β_1 and β_2.

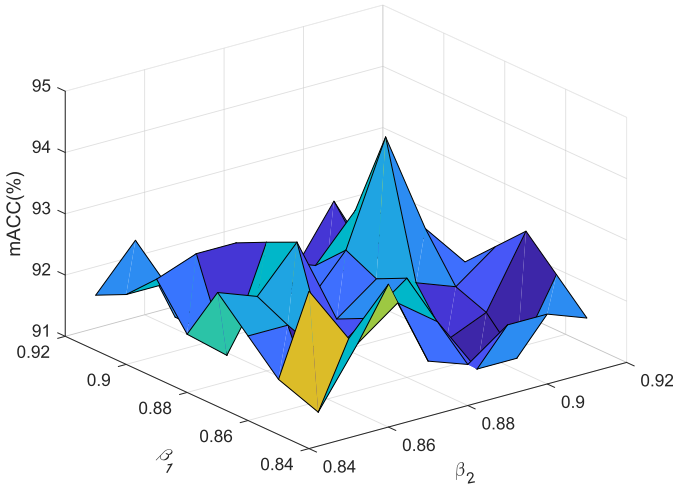


Fig. 12. MACC versus the hyperparameter values of β_1 and β_2 for UC-Merced.

V. Conclusion

In this article, we propose a multiscale incremental learning method for RSSC, aiming to alleviate the stability-plasticity dilemma, i.e., balancing the learning of new tasks and maintaining the performance of old tasks. Orienting to the characteristics of remote sensing scenes with large scale changes, we propose a multiscale feature extraction network as the backbone network, in which MSDW Conv strategy is designed to extract multiscale features to increase the information interaction between different feature groups and enlarge the receptive fields in a multiscale manner and ECA modules to enhance the model's ability to focus on key information. In the incremental learning scheme, new modules are continuously generated to fit the residuals between the labels and the old model output as new tasks come in, maintaining the performance of the old tasks while enhancing the model's plasticity for new tasks. To avoid continuous expansion of the model with the addition of new tasks, we compress the model to remove redundant parameters and reduce feature dimensions. We compare our method with some state-of-the-art methods on three datasets with plenty of experiments, especially for ablation experiments,

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to illustrate the effectiveness of each part of the proposed network.

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